3. Sequences and series

What students need to learn:

Sequences, including those given by a formula for the nth term and those generated by a simple relation of the form $x_{n+1} = \mathrm{f}(x_n)$.	
Arithmetic series, including the formula for the sum of the first n natural numbers.	The general term and the sum to n terms of the series are required. The proof of the sum formula should be known. Understanding of Σ notation will be expected.
The sum of a finite geometric series; the sum to infinity of a convergent geometric series, including the use of $\lvert r \rvert < 1$.	The general term and the sum to n terms are required. The proof of the sum formula should be known.
Binomial expansion of $(1 + x)^n$ for positive integer n. The notations $n!$ and $\binom{n}{r}$.	Expansion of $(a + bx)^n$ may be required.